

Derbyshire Shared Care Pathology Guidelines

Management of children with egg allergy and referral to paediatric allergy clinic

Purpose of Guideline

To give guidance on how to identify those children who have egg allergy or suspected egg allergy. To outline the management of these children and young people.

To outline the process and indication for referral to paediatric allergy clinic. To outline the indications for prescription of adrenaline auto injectors.

The management of anaphylaxis is **not** outlined here – **Please refer to [appendix 2](#) or your usual emergency pathway**

Background

Egg allergy may be defined as an adverse reaction of an immunological nature induced by egg protein. It is extremely common with at least 2% of the childhood population suffering from egg allergy and 0.1% of adults. It is usually IgE-mediated, but occasionally children will present with non-IgE mediated allergy. Egg allergy presents most commonly in infancy, often after the first apparent ingestion with rapid onset of urticaria and angio-oedema; severe reactions involving airway narrowing are uncommon. The clinical diagnosis is made by the combination of a typical history of urticaria and/or angio-oedema/vomiting/wheeze with rapid onset (usually within minutes) after ingestion of egg.

History

The most important part in diagnosis of egg allergy is the history. This requires taking an **Allergy focused history**: Please ask for occurrence of all of these symptoms.

IgE mediated (acute onset)	Non IgE mediated(Delayed onset)
Skin • Pruritis • Erythema • Acute urticaria (localised/ generalised) • Acute angioedema (face, eyes, lips) Gastrointestinal • Angioedema of lips/tongue/palate • Oral pruritis • Colicky abdominal pain • Vomiting and diarrhoea Respiratory • Upper respiratory (nasal itching, sneezing, rhinorrhoea, congestion) • Lower respiratory (cough, wheeze, chest tightness, shortness of breath)	Skin • Pruritis and erythema • Atopic eczema Gastrointestinal • Gastrooesophageal reflux disease • Loose/frequent stools • Blood and/or mucus in stools • Abdominal pain • Infantile colic • Constipation • Food refusal or aversion • Peri-anal redness • Pallor/ tiredness • Faltering growth (with one or more of above)

The timings of any reactions or symptoms need to be recorded in relation to possible egg exposure. The following should also be identified:

Identify allergen - Quantity of egg ingested, type of egg- baked, loosely cooked (egg scrambled, fried) or raw (eg fresh mayonnaise or cake mix)
- First exposure? previous exposure?
- Known allergen, other allergies

Identify any other triggers/ co-factors - temperature, exercise induced, intercurrent illness.

Identify atopic tendency – Asthma/wheeze, eczema, hayfever

Identify medications – antihistamine, steroids, beta 2 antagonists, inhaled medication

Identify strong family history – parents, first degree relatives with asthma, hay fever, food allergies, other known allergies.

Medications and Acute Management

Identify and treat anaphylaxis (see [appendix 2](#) - '*The management of anaphylaxis*')

If no symptoms of anaphylaxis treat symptoms as appropriate:

Oral antihistamines eg Chlorphenamine (as per BNFc)

Initial dose, can be repeated up to x 4 in 24hours if required

Salbutamol – 5 – 10 puffs of MDI via spacer, repeat as required in response to treatment

Oral corticosteroids eg prednisolone

There is no good evidence to support routine steroid administration in allergic reactions.

A 3-day course of oral prednisolone can be considered in the following circumstances;

- Acute exacerbation of wheeze in known asthmatics
- Ongoing and troublesome urticaria/ angioedema
- Ongoing symptoms not responding to regular antihistamines

Investigations

No investigations or allergy testing is required in the acute phase or management of an allergic reaction to food.

Children do not need to be subjected to Specific IgE testing looking for confirmation of egg allergy if it is very clear and evident from the history. Children can be referred to the paediatric allergy clinic for Skin Prick Tests +/- Specific IgE tests if this is required to help make a diagnosis or if the cause of the allergic reaction is not clear.

Criteria for referral to paediatric allergy clinic:

Most children who present only with cutaneous symptoms or mild gastrointestinal symptoms can safely be managed in primary care setting without onward referral to allergy clinic.

Based on the allergy-focused clinical history, BSACI (British Society of Allergy and Clinical Immunology) recommends that referral to secondary or specialist care be considered in any of the following circumstances in children with Egg allergy:

- Children with Anaphylaxis to egg – demonstrated by allergy symptoms that affected:
 - Breathing (cough, wheeze or swelling of the throat, e.g. choking)
 - Gut (**severe** vomiting or diarrhoea)

- Circulation (faintness, floppiness or shock)
- Children who also receive regular asthma preventative treatment and/or have poorly controlled asthma
- Where diagnosis is not clear and needs to be confirmed or excluded
- Severe eczema in children on an egg-containing diet, where control of eczema cannot be gained despite optimised treatment and egg appears to be a trigger. (*These patients will usually already be under the care of dermatology*)
- Persistent egg allergy after age 6-8 years
- Egg allergy with requirement for yellow fever immunization
- Significant reactions to trace amounts of egg
- Egg allergy with another major food allergy (therefore multiple food allergies)

All children who fit the above criteria should have a referral to paediatric allergy clinic. This can be accessed via choose and book pathway.

Management advice and Follow up

When observations are stable and the patient has shown a good response to treatment and if there were no signs of anaphylaxis, the patient may be discharged home with advice on the following:

- Regular anti histamines for up to 3 days if required. Seek review if symptoms persist beyond this.
- Avoid egg in diet for 6 -12 months after initial reaction.
- Verbal and written advice should be given regarding egg avoidance. Information leaflets are available for many allergies including egg and good quality patient information can also be obtained from www.allergyuk.org and anaphylaxis campaign: <https://www.allergyuk.org/resources/egg-allergy-factsheet/>

Patients **MUST** be provided with a written management plan, please access these with the following link:

<http://www.bsaci.org/about/pag-allergy-action-plans-for-children>

- Patients should be provided with an egg ladder so that when re-introduction is advised they are able to follow a structured plan; the egg ladder is attached in [Appendix 1](#)

Advice on re-introduction of Egg

The natural history of egg allergy in children is that resolves spontaneously in the majority. Various studies given different predictions varying from 90% at age 2 to 66% at age 5 years. Parents should be reassured that resolution by age 2 is common and by school age is even more so.

Resolution of egg allergy occurs in stages starting with tolerance to well-cooked egg (e.g. cake), then lightly cooked egg (e.g. scrambled) followed finally by raw egg. Therefore, children who tolerate cooked egg may still react to raw or undercooked egg. As a rule, reactions do not become more severe over time and often become less severe.

The speed with which egg allergy resolves can vary greatly between individuals, and therefore the timing and appropriateness of reintroduction should be individually assessed. Reintroduction should not be attempted within 6 months of a significant reaction to egg. Children who have had only mild symptoms (only cutaneous symptoms) on significant exposure (e.g. a mouthful of scrambled eggs) with no ongoing asthma may have well-cooked egg (e.g. sponge cake) introduced from the age of about 12 months – 2 years at home. If this is tolerated then reintroduction of lightly cooked egg can be done as per the egg ladder ([Appendix 1](#)). If there is a reaction at any stage, the previously tolerated diet should be continued and further escalation considered after another 3 - 6 months.

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Reintroduction at home should not be attempted if there have been significant gastrointestinal, respiratory or cardiovascular symptoms during previous reaction; if only a trace amount has ever been ingested or there is ongoing asthma/ wheezing symptoms. These children should be referred to paediatric allergy clinic for further review and assessment before re-introduction.

Egg Allergy and Immunisations

Measles, mumps and rubella vaccine. All children with egg allergy should receive their normal childhood immunizations, including the MMR vaccination as a routine procedure performed by their family doctor/nurse. This advice should be provided at diagnosis. Studies on large numbers of egg-allergic children show there is no increased risk of severe allergic reactions to the vaccines.

Influenza vaccine.

Nasal flu vaccine - Recent studies have demonstrated that children with egg allergy can safely be vaccinated with the nasal flu vaccine (Fluenz Tetra) in any setting (including primary care and schools). This includes children with previous anaphylaxis to egg. The only contraindication is those children with anaphylaxis so severe they were admitted to PICU. (See [Appendix 3](#) for more information)

Inactivated (injected) influenza vaccines that are egg free or have an ovalbumin content <0.12 µg/ml may be used safely in individuals with egg allergy, in primary care. Those vaccines with higher ovalbumin content should be avoided.

Further information can be found in the DoH Green Book, available at <https://www.gov.uk/government/publications/influenza-the-green-book-chapter-19>

Yellow fever vaccine

Yellow fever vaccine is contraindicated in children with egg allergy. For those children requiring this vaccination, please refer to allergy clinic for further assessment.

Prescription of Adrenaline Auto Injectors “AAI” (eg JEXT/ Epipen)

All children who have had anaphylaxis to egg should be prescribed AAI before discharge with training in how and when to use them.

The following guidance on other indications for AAI prescription is taken from EAACI (European Academy Allergy and Clinical Immunology)

Absolute indications:

- Previous anaphylaxis triggered by egg
- Co-existing unstable or moderate to severe, persistent asthma and symptoms of IgE mediated allergy

Relative indications:

Egg allergy and **2** or more of following risk factors:

- Older age > 12yrs
- Patient is often/frequently remote from medical help
- Allergic reaction to very small amounts/ traces
- Asthma (on regular preventer)

Patients fulfilling the above criteria should have an Adrenaline Auto Injector (AAI) prescribed. They should be discussed with a senior before discharge and prescription of AAI.

Appendix 1 – Egg ladder for re-introduction



Derbyshire Children's Hospital How do I introduce egg into my child's diet?

Most children with egg allergy grow out of it in early life. Children will start to tolerate well-cooked egg first (foods containing cooked egg such as cake) followed by lightly cooked whole egg (e.g. scrambled egg) then finally uncooked whole egg.

A medical professional has recommended that you begin to introduce egg into your child's diet at home. We do this in a stepwise approach.

The table below shows examples of foods containing well cooked egg, loosely cooked egg and raw egg.

Well cooked egg	Loosely cooked egg	Raw egg/ Undercooked egg
Cakes	Scrambled egg	Homemade mayonnaise
Biscuits	Boiled egg	Raw cake mix
Dried Egg Pasta	Omelette	Some sorbets containing egg
Well-cooked fresh egg pasta	Poached egg	Fresh Mousses
Dried egg noodles	Fried egg	
Shop bought pancakes/Yorkshire puddings	Homemade pancakes/Yorkshire puddings	Certain homemade sauces- tartare, horseradish
Quorn TM or similar meat alternative containing egg	Fresh egg pasta	Fresh Ice cream containing egg
Cooked meat dishes e.g burgers and sausages containing eggs	Quiche/flan/Spanish tortilla	
Shop bought meringue	Homemade meringue	

Using the egg ladder

This egg ladder explains how to perform the egg challenge at home. Children who have had more severe symptoms may need to have a challenge performed under hospital supervision.

The dietitian or your doctor will advise when it is appropriate to try each stage of reintroduction. The following information is only a guide and may need to be adapted for each child.

For each stage, use one food from the above list and gradually introduce starting with only a small amount and build up over 1-2 weeks.

If a reaction occurs, stop re-introducing the food and return to the previously tolerated stage. Stay at this stage and re try again in 1-2 months.

If your child has been prescribed antihistamines please make sure they are with you when trying new foods from the egg ladder.

If your child has never had any baked egg products, start with stage 1. However, if they had exposure to egg in some form by accident and tolerated this with no reaction, then start with the stage this food is in.

Having a diary of amounts of food introduced and any symptoms experienced maybe useful

Be aware that

- The amount of egg, the temperature and length of cooking can all affect the how much the egg protein is cooked
- Raw or undercooked eggs are more likely to cause a reaction than well cooked egg
- It is also possible to react to raw or undercooked egg even if a softly cooked whole egg has been previously tolerated

Well-cooked egg (Stage 1)

- Postpone the reintroduction if your child is unwell and have oral antihistamines available
- Bake or buy a cake containing egg, ensure that the other ingredients of the cake are tolerated
- Begin by rubbing a small amount of cake on the inner part of your child's lips. Wait for 30min but carry on with normal daily activities
- If there have been no symptoms, give your child a pea-sized amount of cake to eat
- A day or two later, if there have been no symptoms; give your child twice the amount of cake to eat
- Increase the amount given until all the cake finished over the 1-2 weeks as tolerated

Well Cooked egg (Stage 2)

Once your child has tolerated cake for a few weeks, you can start to introduce other well cooked egg products into their diet, see table above for examples.

Introduce these products one at a time to ensure they are tolerated well.

Once your child has tolerated well cooked eggs in different forms for a number of weeks you can move onto stage 3.

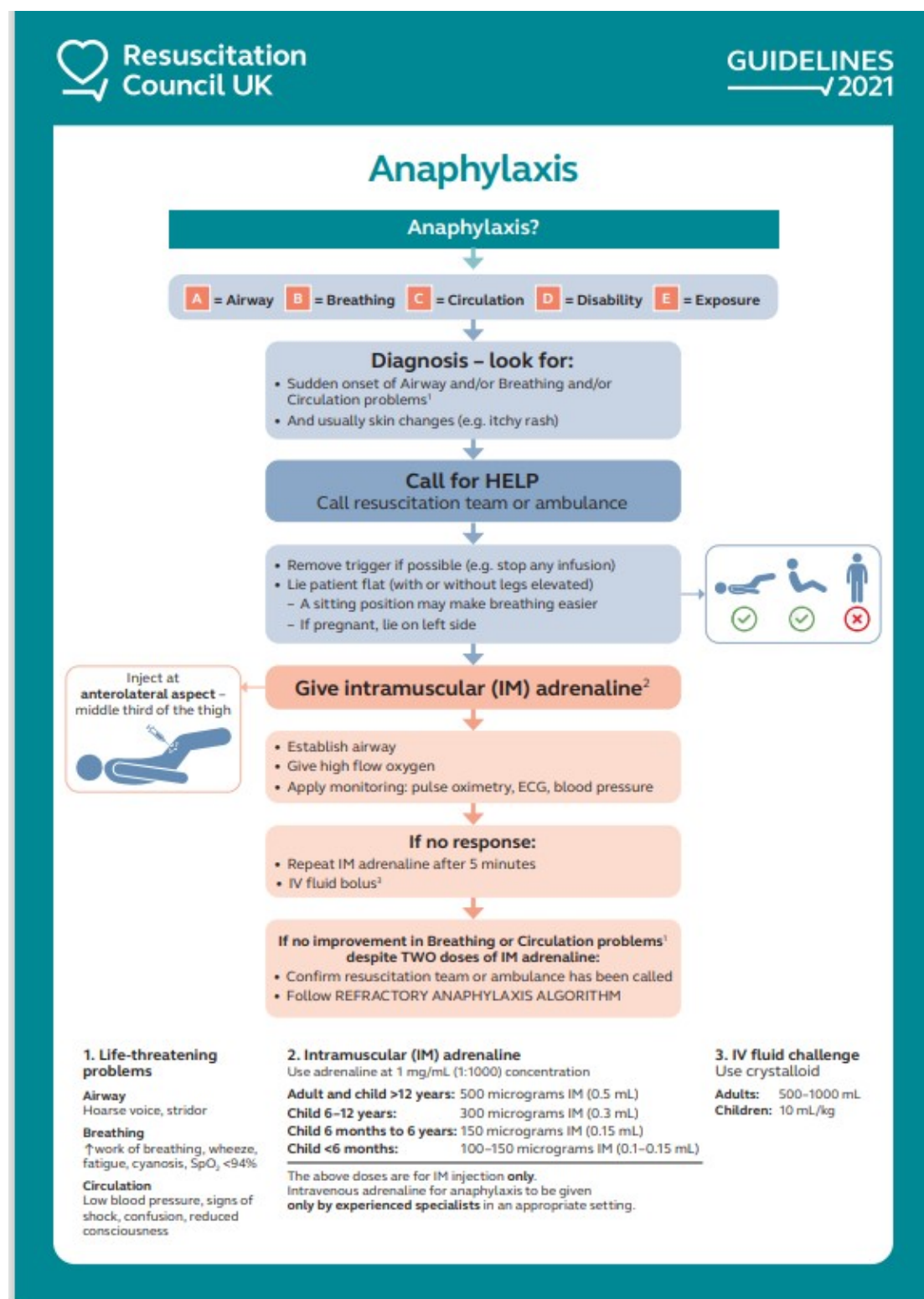
Loosely cooked egg (Stage 3)

- Cook a portion of scrambled eggs (well cooked)
- Give your child a small amount to eat (e.g. ¼ teaspoon). If this is well tolerated then gradually increase the amount over 1-2 weeks (e.g. ½ teaspoon, 1 fork full, whole scrambled egg)
- If it is well tolerated try reducing the cooking time of the scrambled egg
- Other loosely cooked egg dishes can then be introduced, see table above for examples
- Once a child can tolerate loosely cooked egg then try and keep in their diet at least 3 x a week

Raw eggs (Stage 4)

There are only a few foods (see above) that contain raw egg. Young children should not be given raw egg due to the risk of food poisoning. If there is a chance your child may be exposed to raw egg (e.g. eating out when abroad) then raw egg can be introduced at home in the same way as the previous steps.

Appendix 2 – Anaphylaxis management



Appendix 3 – Immunisation information

[Influenza vaccine and egg allergy - BSACI](#)



British Society for Allergy and Clinical Immunology
Paediatric Committee

2015 / 16 Influenza vaccine recommendations for children with egg allergy

During the 2015 / 16 influenza season, the nasal influenza vaccine (Fluenz Tetra) will be offered to children aged 2-4 years old and schoolchildren in years 1 and 2. Primary school-aged children in areas of the country that previously participated in the pilot programme last year will also be offered the vaccine. Schoolchildren will be offered the vaccine via a schools-based programme.

Children aged 2 -18 years with long term health conditions (e.g. asthma requiring previous hospital admission), should also be offered Fluenz Tetra. Children between 6 months and 2 years with long-term health conditions will be offered the injected flu vaccine.

In all settings providing vaccination, facilities should be available and staff trained to recognise and treat anaphylaxis

The Joint Committee on Vaccination and Immunisation have advised that **children with egg allergy can safely be vaccinated with Fluenz Tetra in any setting (including primary care and schools)**. This includes children with previous anaphylaxis to egg. This advice is based on the SNIFFLE studies, in which 887 egg-allergic children safely received Fluenz vaccine, of whom over one third had a history of previous anaphylaxis to egg.

The only exception is for children who have previously had an anaphylaxis to egg so severe that they required ventilation on an intensive care unit as a result. These children are likely to tolerate Fluenz, but were excluded from the SNIFFLE studies. Therefore, these children should be referred to a specialist for immunisation in hospital.

Fluenz Tetra should not be administered to a child with current or recent acute wheezing in the 72 hours preceding vaccination, or who have required oral steroids in the previous 2 weeks.

Inactivated (injected) influenza vaccines that are egg free or have an ovalbumin content < 0.12 µg/ml may be used safely in individuals with egg allergy, in primary care.

Further information can be found in the DoH Green Book, available at:

https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/427809/Green_Book_Chapter_19_v9_0_May_2015_.PDF

References

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2. Anaphylaxis: assessment and referral after emergency treatment. NICE CG134: (Aug 2020). <https://www.nice.org.uk/guidance/cg134>
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